

DATS 598: Data Science Capstone
Journal Entries and Progress Documentation

Statistical Truth Detection System

Eric Crisp
University of Pennsylvania
ecrisp@upenn.edu

Thursday 2nd October, 2025

Contents

Journal Entry - Week 4 - Statistical Truth Detection System	2
--	----------

Journal Entry - Week 4 - Statistical Truth Detection System

Date: Thursday 2nd October, 2025

Duration: 10 hours

Objective

Complete initial data investigation, prototyping of architecture and product format, begin implementation.

Activities Completed

- Continued deep dive into the NLP processes needed, finding open source libraries where applicable, and determine the necessary, nice to have for project deliverables.
-

Key Findings

Next Steps

- Pull in data sources for fact-checking (e.g., FEVER, HuggingFace, Wikidata, etc.).
- Begin building out the backend applications via notebook (sentence parsing and analysis, API hooks, open source and python compability and configurations for environment stability).
- Begin prototyping the web app and setting up the tech stack (confirmation of CI/CD pipelines, containerized code, frontend-backend communication).

The current tech stack (as planned, and currently built is this):

- Backend: Python, FastAPI, Uvicorn
- Frontend: JavaScript, React, Vite
- VCS/Deployment: Git, GitHub Actions for CI/CD, AWS for hosting (?)
- Database: PostgreSQL, if RAG (?)
- NLP: SpaCy, NLTK, HuggingFace Transformers
- Containerization: Docker (Docker compose)
- EDA Packages: Jupyter Notebooks, Pandas, Matplotlib, Seaborn, NLTK, SpaCy, Transformers, Torch